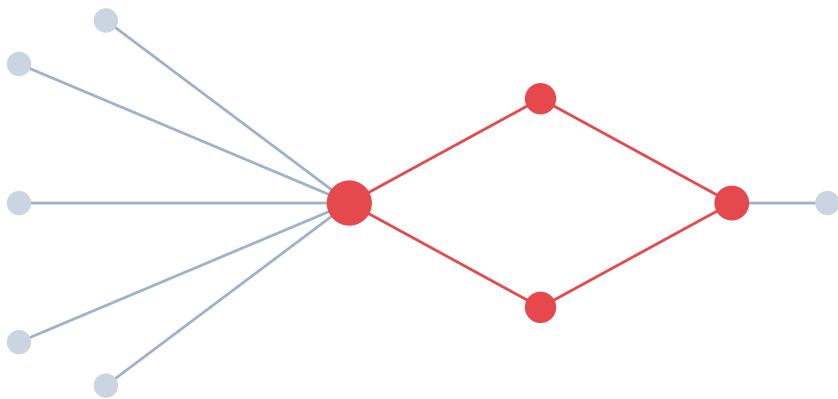


PROJECT REPORT

Cygnus AI

Finding money-mule networks in mobile financial services by following the money through a graph, with an explainable AI risk score for every account.

"Don't just flag transactions. Follow the money."



TEAM MEMBER	ROLE	UNIVERSITY
Sadekunnahar Marufa	Platform & integration (backend, dashboard)	Jahangirnagar University
AI Amin	Graph algorithms & pattern detection	East West University
Tanvir Hossain Shihab	Machine learning, risk scoring & security	Jahangirnagar University

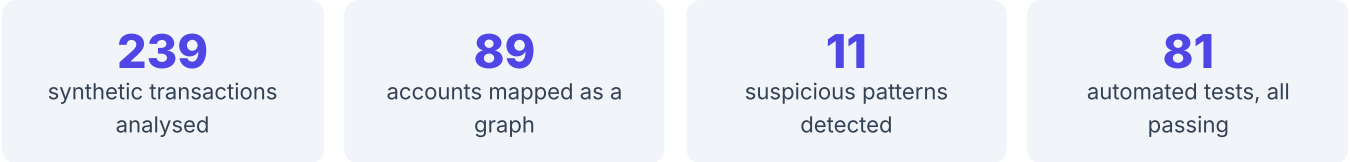
Source code: github.com/ManamiMayoki/CPC_DIU_HACKATHON_2026

Live demo: <http://20.219.7.216:3001>

Date: 7 October 2026

1 Summary in one page

Fraud in mobile financial services (MFS) rarely looks like one bad transaction. Criminals spread money across many “mule” wallets so that every single transfer looks normal. The fraud only becomes visible when you look at **how accounts are connected**. Cygnus AI turns transactions into a network, finds the suspicious shapes money leaves behind, scores every account from 0 to 100 with machine learning, and tells an investigator in plain words **why** an account was flagged and **what to check next**.



Problem	For upay’s AML and fraud analysts , money-mule rings, digital hundi and stolen-phone account takeovers hide in normal-looking wallet transfers, because rule-based checks look at one transaction at a time. On our labelled benchmark, rules miss 686 of 1,010 suspicious accounts and raise 904 false alarms . We built an AI product that uses the transaction graph to rank and explain risky wallets, with success measured by precision, recall, PR-AUC and alerts per true case .
Idea	Treat accounts as nodes and transfers as arrows between them. Suspicious behaviour has a recognisable shape in that network.
What we built	A working end-to-end system: data validation, a transaction graph, 11 pattern detectors, an Isolation Forest anomaly model, an explainable 0–100 risk score, an AI Investigator that writes a case briefing with next steps, and an interactive dashboard where you can “Follow the Money” hop by hop.
AI approach	Graph analytics (11 detectors, time-ordered flow tracing, 2-hop neighbourhood features), unsupervised anomaly detection (Isolation Forest) and a supervised gradient-boosting model, combined into one transparent score: 40% ML + 40% graph + 20% behaviour . On 423,042 held-out transactions the best model reaches precision 94.8%, recall 83.8%, PR-AUC 0.943 (rules: 0.201). Every score comes with evidence built from measured numbers, and an optional Claude (LLM) narrative rewrites the briefing in plain language.
What happened, why risky, what next	Every alert answers the guideline’s three questions: the graph shows what happened to the money, the evidence and score show why it is risky , and the AI Investigator lists what upay should do next (verify NID/device/SIM, check agent cash-out, step-up authentication, escalate, draft an STR).
Impact	Measured on the benchmark: 73.6% fewer alerts to find the same suspicious accounts as rules, false alarms 904 → 46 , and analyst time per true case 76 → 21 minutes (at an assumed 20 minutes per alert). For upay this means catching mule networks before cash-out and protecting customers whose phones are stolen, without bothering legitimate users.

How this report maps to the judging criteria

CRITERION (WEIGHT)	WHERE TO LOOK
Problem relevance (20%)	Sections 3 and 7: Bangladesh MFS figures from public sources, upay location-risk and hundi detectors, why rules miss them
AI/ML depth (20%)	Sections 8 and 12: models, labelled benchmark, precision / recall / F1 / PR-AUC, feature importance
Business / customer impact (20%)	Section 15: measured alert reduction, modelled savings, pilot design, business model
Prototype quality (15%)	Sections 9, 11 and 13: working dashboard, analyst workflow and case management, tests
Innovation (10%)	Sections 4, 7 and 12: network view, and what graph intelligence catches that rules miss

Scalability & integration (10%)	Sections 10 and 14: service APIs, throughput benchmark, streaming and graph-database architecture
Responsible AI & security (5%)	Sections 11.5 and 14: access control, audit log, PII masking, bias check, governance roadmap

2 Built for upay: what Cygnus AI delivers

We designed Cygnus AI around what upay's fraud, AML and compliance teams need: Bangladesh-specific risks, measured model performance, a complete analyst workflow, concrete security controls and a credible path to upay scale. This table maps each evaluation criterion to what we built and where it is shown. Anything that needs real upay data or a live pilot is stated as roadmap, not as a result.

CRITERION	WEIGHT	WHAT UPAY NEEDS	WHAT CYGNUS AI DELIVERS	SECTION
Problem relevance	20%	Bangladesh MFS evidence and local risk scenarios, including agent hundi and stolen-phone takeover	Public Bangladesh MFS figures, plus two upay detectors: risk-area account takeover and agent hundi	3 and 7
AI/ML depth	20%	Measured precision, recall, F1, PR-AUC and false positives; model comparison; feature importance; temporal and graph learning	8 models compared on 423k held-out transactions: best F1 0.89, PR-AUC 0.94, false positives 0.12%	12
Business impact	20%	Fewer alerts, less analyst time per case, lower false-positive cost, and a pilot to prove it	73.6% fewer alerts at equal recall, false alarms 904 → 46, and an 8-week upay pilot to measure real savings	15
Prototype quality	15%	A working end-to-end product with a full analyst workflow and case management	Working analyst flow: sign in, alert, trace, investigate, open case, escalate, STR/SAR export, audit trail	13
Innovation	10%	Proof that graph intelligence finds what conventional rules miss	Rules miss 686 of 1,010 suspicious accounts; the graph model catches 561 of those	12
Scalability	10%	High-volume throughput, streaming ingestion, graph database and integration with upay systems	1.06M transactions scored in 18.9 s on one laptop process; API p95 under 6 ms; streaming + graph-DB path	14
Responsible AI	5%	Access control, audit logging, privacy, bias testing and governance	Two roles, hash-chained audit log, PII masking with logged reveal, bias check by account tier and district	14

3 The problem

Bangladesh has tens of millions of mobile wallet users, and digital money now moves salaries, remittances, shop payments and person-to-person transfers. That convenience is also used by fraudsters. A common pattern is the **money-mule network**: stolen or illicit money is split across many wallets (often belonging to people who were paid or tricked into "renting" their account), moved quickly through several hops, and then cashed out.

Why today's checks miss it

- **Each transfer looks normal.** Five payments of BDT 9,800 from five different people pass every per-transaction rule.
- **The pattern is in the connections.** Only when you see that all five go to the same collector, who forwards the money within minutes, does it look suspicious.
- **Investigations are slow.** Analysts trace flows account by account in flat tables. One multi-hop case can take hours, and by then the money is gone.
- **Alerts without reasons.** Black-box scores and threshold alerts are hard to act on and hard to defend to auditors and regulators.
- **No labelled data.** Real fraud labels are rare and sensitive, so supervised models are hard to train.

What a rule sees

A-214 → ? BDT 9,800 ✓
A-377 → ? BDT 9,650 ✓
A-108 → ? BDT 9,900 ✓
A-652 → ? BDT 9,700 ✓

What the graph sees

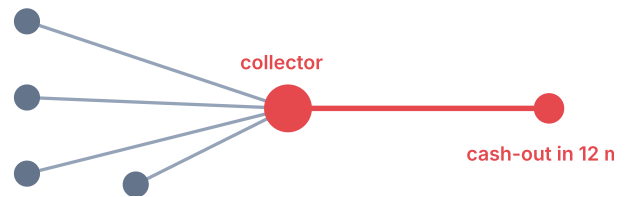


Illustration: four "clean" transfers are really one fan-in collector that passes the money on.

Our demo data includes a realistic version of this scheme, the **MFS mule ring**: eight feeder wallets send deposits just under BDT 10,000 to a hub, the hub forwards each one to four mule wallets within one to three minutes, the mules pay a cash-out agent, and 90% of the money flows back to the hub. Cygnus AI scores the hub **92.3/100 (CRITICAL)**.

The problem in numbers (public sources)

239.3M

MFS accounts in
Bangladesh, January
2025 [1]

~22M

MFS transfers a day:
678.6M worth Tk 1.58
trillion in October 2025
[2]

30,199

suspicious transaction /
activity reports to BFIU in
FY2024-25, up 74% [3]

**Tk 75,000
cr**

moved by digital hundi in
one year through about
5,000 agents of bKash,
Nagad, upay and Rocket
(CID) [4]

BFIU also suspended **21,725 MFS accounts** in nine months of 2023 for online gambling and hundi [5]. At around 22 million transfers a day, no team can review rows; the suspicious network has to surface itself. Our synthetic data plants the Bangladesh typologies behind these figures:

- **Digital hundi via agents:** an agent wallet funded every month pays the same local wallets
- **Gambling collection:** hundreds of small deposits into a few collector wallets
- **Mule rings:** recruited wallets forward money within minutes and cash out
- **Structuring:** repeated transfers just under a reporting threshold
- **Circular layering:** money loops through 3 to 6 wallets back to its origin
- **Stolen-device takeover:** a wallet suddenly used from an unfamiliar risk area on a new device

Hackathon track. This project belongs to **Track 01: Trust & Risk Intelligence** ("fraud detection, scam intelligence, account takeover, mule networks, agent risk"). Trust is the foundation of wallet adoption: if users fear losing money, they stop using digital finance.

4 Our idea: look at the network, not the row

Instead of checking transactions one by one, Cygnus AI draws them as a **graph**. Every wallet is a dot (a *node*), and every transfer is an arrow (an *edge*) from sender to receiver that remembers the amount and the time. Laundering schemes leave recognisable shapes in this picture: a star of arrows pointing into one account, money that comes back to where it started, a long chain of quick hops. We detect those shapes with graph algorithms, measure how unusual each account is with machine learning, and combine everything into one score an investigator can understand.

In one sentence: accounts become nodes, transactions become edges, and suspicious shapes become evidence.

Three things make this different from a normal rule engine:

1. **Network view.** Patterns that span many accounts (rings, chains, collectors) are visible at once.
2. **No labels needed.** The anomaly model learns what “normal” looks like from the data itself, so it can work on day one at a new provider.
3. **Explainable by design.** Every score is split into its parts and comes with plain-language evidence built only from measured numbers. Nothing is invented.

5 How it works, step by step

1 · Validate

Reject bad records: missing fields, zero/negative amounts, NaN, values above 10^9 , bad timestamps, self-transfers, duplicates.

2 · Build graph

NetworkX directed graph. Nodes are accounts; edges hold amount, time and transaction ID.

3 · Detect

11 detectors scan the graph for suspicious shapes (Section 5).

4 · Featurise

Per account: graph features (degrees, cycles, chain length, risky neighbours) and behaviour features (velocity, pass-through, volatility).

5 · Score

Isolation Forest finds outliers. A composite 0–100 score adds graph and behaviour signals.

6 · Explain

Evidence, risk tier and an AI Investigator briefing (typology, findings, next steps), shown in the dashboard.

The whole intelligence layer is one Python call, `run_pipeline(transactions)`, which returns a structured JSON result (documented in `docs/api-contract.md`). The backend runs it on the sample data at start-up and again whenever a user uploads their own transactions.

Input format

Any transaction feed with these five fields can be analysed. This is the only thing an MFS provider would need to map from their own system.

FIELD	MEANING	RULES ENFORCED
transaction_id	Unique ID (generated if missing)	Must be unique in the batch
sender_id	Wallet that sends money	Required, cannot equal receiver
receiver_id	Wallet that receives money	Required
amount	Transfer amount	Number, greater than 0, finite, below 1,000,000,000
timestamp	When it happened	ISO-8601 text or Unix time; no time zone is read as UTC

6 The suspicious patterns we detect

Each detector is a small, readable graph algorithm with clear thresholds. A detector firing is a **signal for review, not proof of crime**. The last column shows how many accounts each detector flagged in the demo dataset.

PATTERN	WHAT IT LOOKS LIKE	WHY CRIMINALS DO IT	FLAGGED
Fan-In	Many accounts send money into one collector account.	Collecting money from mules before it is cashed out.	3
Fan-Out	One account sprays money out to many receivers.	Splitting stolen or illicit funds across many wallets.	2
Rapid Movement	An account receives money and forwards most of it within an hour.	Pass-through (mule) accounts that never keep the money.	13
Transaction Chain	Money hops through three or more accounts in a row.	Putting distance between the source and the final receiver.	19
Circular Flow	Money travels A → B → C and back to A.	Layering: moving money in loops to hide where it came from.	9
Structured Transactions	Many transfers kept just under a reporting or limit threshold.	Smurfing: breaking a large sum into small, 'safe-looking' pieces.	4
Hundi Operator	An agent or wallet takes large informal transfers from a few funders and pays the same beneficiaries cycle after cycle.	Digital hundi: paying out remittance that came in outside the legal channel.	1
Hundi Funder	An account that repeatedly sends large transfers to a suspected hundi operator.	The source side of a hundi network.	2
Risk-Area Takeover	A wallet is suddenly used from a listed risk area outside the customer's own base, with other takeover signals.	Stolen phones gathered in one area and used to drain wallets.	3
Takeover Collector	Collects money from several wallets during suspected account takeovers.	The account that receives drained funds.	1
Coordinated Network	A tight group of accounts that mostly pay each other.	A ring of accounts controlled by the same people.	4

How the detectors decide

- **Fan-in / fan-out:** at least 4 distinct counterparties on one side, at least twice as many as on the other side, arriving (or leaving) within a 2-hour burst.
- **Rapid movement:** an inflow of at least 1,000, of which 70% to 110% is forwarded to another account within 1 hour.
- **Transaction chain:** a directed path of 3 or more hops through intermediary accounts.
- **Circular flow (time-ordered):** a loop of 3 to 6 accounts where every hop happens after the money arrived, the loop closes within 6 hours, and 50% to 110% of the value comes back. Random peer-to-peer loops are not flagged.
- **Structuring:** 3 or more transfers between 85% and 100% of a 10,000 threshold within 24 hours. Both the splitting account and the collecting account are flagged. (The 10,000 value is a configurable example, not an official limit.)
- **Coordinated network:** a strongly connected group of 4 or more accounts with internal link density of at least 0.4.

7 Two detectors built for upay and Bangladesh

Two Bangladesh-specific risks need more than money-flow shape. **Stolen-device takeover**: phones stolen in many places are gathered in a few locations and used to drain wallets. **Digital hundi**: CID found Tk 75,000 crore moved in one year through about 5,000 MFS agents, including upay agents [4]. Both detectors score a wallet against **its own normal behaviour**, so ordinary customers are not bothered.

Risk-area account takeover

Risk judged against each customer's own baseline

- Learns each wallet's usual areas and handsets from its own history
- Risk areas come from upay's fraud team (demo uses placeholders Zone-13, Zone-14)
- Alert needs at least two signals: area new for this customer, new handset, unusually fast outflow, new receivers
- Also flags the collector wallet that gathers money from several taken-over accounts

No hassle for legit users. Customers who regularly transact from a risk zone are not flagged. A first-time hit triggers step-up checks (PIN / OTP) or analyst review, never an automatic block.

Measured: Precision 93.1%, recall 84.4%, false-positive rate 0.02%. Residents of risk areas flagged: 0 of 5,136. Visitors: 8 of 1,027. Collector wallets: 24 of 24.

Agent hundi operator

Recurring foreign-linked money through one agent

- Agent funded informally (no licensed inward-remittance or distributor source)
- Pays out to five or more beneficiaries soon after funding
- The same beneficiaries are paid again in two or more cycles
- Also flags the funders who keep topping the agent up

No hassle for legit users. Agents doing licensed remittance payout are not flagged (0 of 117), and families who simply receive money are scored on their own history.

Measured: Operator: precision 100%, recall 91.7%. Funders: 100% / 92.9%. Licensed remittance agents flagged: 0 of 117.

Detector	True cases	Flagged	Precision	Recall	False-positive rate
Risk-area account takeover (customer at risk)	128	116	93.1%	84.4%	0.020%
Takeover collector	24	24	100.0%	100.0%	0.000%
Hundi operator	24	22	100.0%	91.7%	0.000%
Hundi funder	56	52	100.0%	92.9%	0.000%

Customers who live in a risk area: 0 of 5,136 flagged. Customers who only visited one: 8 of 1,027. Agents doing licensed remittance payout: 0 of 117. Risk areas in the demo are placeholders (Zone-13, Zone-14) that upay's fraud team would replace with its own list. The simulated cases are clean, so these figures are an upper bound.

Action, not punishment. A location alert asks for step-up verification (PIN or OTP) or puts the transfer in an analyst queue. A hundi alert opens a case for the AML team with the agent, the monthly cycle and the recipient wallets, ready for an STR to BFIU if confirmed. Neither blocks an account automatically.

8 The AI approach

6.1 Anomaly detection without labels

Real fraud labels are scarce, so we use **Isolation Forest** (scikit-learn), an unsupervised model. It builds many random trees that split the data; accounts that can be isolated in very few splits are unusual. Each account is described by a combined feature vector:

Graph features

in/out degree, weighted in/out flow, net flow, unique counterparties, fan-in and fan-out scores, number of cycles, chain length, network size, number of suspicious neighbours.

Behaviour features

transaction count, total in/out, average, maximum and spread of amounts, transactions per active hour (velocity), in/out ratio, counterparty diversity, pass-through ratio, amount volatility, flow imbalance.

Features are scaled with a `RobustScaler` so a single extreme account cannot distort the rest. The raw model output is mapped to a 0–100 anomaly score: normal accounts fall roughly in 5–38, outliers in 50–98. A fixed random seed makes every run give the same result.

6.2 One explainable risk score

$$\text{Risk} = 0.40 \times \text{ML anomaly} + 0.40 \times \text{Graph pattern} + 0.20 \times \text{Behaviour}$$

PART	WEIGHT	WHAT IT MEASURES
ML anomaly	40%	How unusual the account is overall, from Isolation Forest.
Graph pattern	40%	Which suspicious patterns the account is part of, and how exposed it is to other flagged accounts.
Behaviour	20%	Velocity, pass-through and flow imbalance, compared with the 75th and 90th percentile of the current population, so thresholds adapt to the data instead of being hard-coded.

Graph points (capped at 100): coordinated network 40, circular flow 35, rapid movement 30, structuring 30, fan-in 25, fan-out 25, transaction chain 20, plus up to 15 for links to other flagged accounts. Patterns that are more typical of laundering weigh more.

The score maps to four tiers: **LOW** 0–40, **MEDIUM** 40–70, **HIGH** 70–90, **CRITICAL** 90–100. Every account record keeps its three parts, so an investigator always sees which signal drove the result.

Worked example, the MFS mule-ring hub: ACC_MULE_HUB **CRITICAL · 92.3**

$0.40 \times 99.8 \text{ (ML)} + 0.40 \times 100 \text{ (graph)} + 0.20 \times 62 \text{ (behaviour)} = 92.3$

Evidence produced by the system:

- Trained classifier rates this account 100% similar to known laundering typologies. Main drivers: total received (BDT) = 85361.6 (typical 1164.65).
- Circular flow pattern detected: funds travelled a 3-account loop (ACC_MULE_01 → ACC_MULE_CASHOUT → ACC_MULE_HUB → ACC_MULE_01) and returned in 81 minutes with 90% of the value.
- Rapid fund movement detected: funds forwarded within under a minute of receipt.
- Fan-in pattern detected: received funds from 9 distinct senders with only 4 outgoing counterparties.
- Structuring detected: sent 8 transactions just under the 10,000 threshold (total 75,009) within 24 hours.
- Transaction chain detected: participates in a multi-hop path spanning 3 hops.
- High transaction velocity: 12.1 transactions per hour.
- High network exposure: directly connected to 5 accounts with flagged patterns.
- High layering passthrough ratio: 87.9% of incoming funds forwarded onward.
- Statistical multivariate anomaly flagged by Isolation Forest (anomaly score: 99.75/100).

6.3 AI Investigator

Finding a risky account is only half the job. The AI Investigator turns the result into a short case an analyst can act on:

- **Names the typology:** for example money-mule collection hub, pass-through mule, layering loop, structuring or coordinated cluster.
- **Explains why** with key findings that quote the measured numbers: how many near-threshold transfers, how fast money was forwarded, how much came back around a loop, the ML outlier score.
- **Recommends next steps written for an MFS investigator:** check whether the wallets share an NID, device or SIM, review agent cash-out records, contact a sample of senders, and for CRITICAL cases escalate the same day, consider a temporary hold under internal policy and prepare a Suspicious Transaction Report (STR) for BFIU.
- **Lists linked accounts** that open in Follow the Money with one click, and **copies a full case report** for the case file.

- **Works fully offline:** the briefing is built by rules from the evidence (`ml/investigator.py`), so it is deterministic and cannot invent facts.
- **Optional Claude narrative:** when an Anthropic API key is set on the backend, the "Generate AI Narrative" button asks Claude to rewrite the briefing as a short investigator narrative, using only the pipeline's facts and a structured JSON output format. Without a key, or on any error, the system falls back to the rule-based briefing and says so.

Real briefing produced for ACC_MULE_HUB: Money-mule collection hub

ACC_MULE_HUB scores 92.3/100 (CRITICAL). It collects split deposits from many wallets and forwards them almost immediately. Detected patterns: fan-in collection, circular flow, multi-hop chain, rapid pass-through, structuring. Received BDT 85,362 and sent BDT 75,009 across 17 transactions.

Key findings

- Sent 8 transfers just under BDT 10,000, totalling BDT 75,009 within 24 hours.
- Received money from 9 different wallets, all within two hours.
- Passed received money straight on 8 times (to 4 wallets), the fastest within under a minute of receipt.
- Money went around the loop ACC_MULE_01 → ACC_MULE_CASHOUT → ACC_MULE_HUB → ACC_MULE_01 in 81 minutes and 90% of it came back.
- Part of a transfer chain spanning 3 hops.
- Passed on 88% of everything it received, keeping almost nothing.
- The Isolation Forest model ranks its overall behaviour as a statistical outlier (ML score 100/100).

Next steps

1. Pull the full list of near-threshold transfers and check whether one person controls the sending wallets (same NID, device or SIM).
2. Compare the combined amount with the account's declared income or business profile.
3. Identify the sending wallets and check how recently they were opened and whether they share a device, SIM or agent.
4. Contact a sample of senders to confirm the purpose of their payments.
5. Review the timing of each receive-and-forward pair; funds held for minutes suggest the wallet is a relay, not the beneficiary.
6. Open linked cases for ACC_MULE_SRC_000, ACC_MULE_SRC_001, ACC_MULE_SRC_002, ACC_MULE_SRC_003 and others in Follow the Money.
7. Escalate to the compliance officer today; consider a temporary hold on outgoing transfers under internal policy and prepare a Suspicious Transaction Report (STR) for BFIU if the review confirms the pattern.

Generated from synthetic data and computed metrics. These are risk indicators for prioritisation, not proof of wrongdoing.

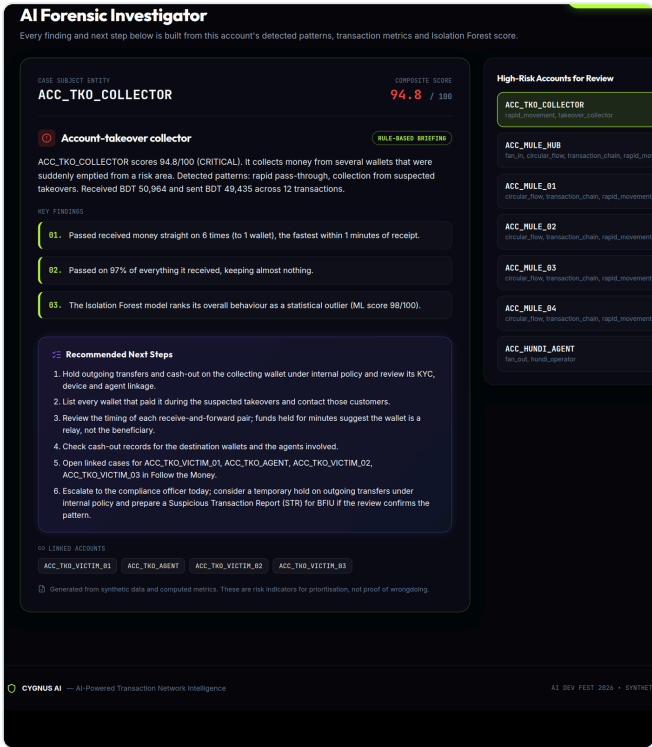
9 The product: what an upay analyst sees

The dashboard is a single web page built for a live demo and for real investigation work.

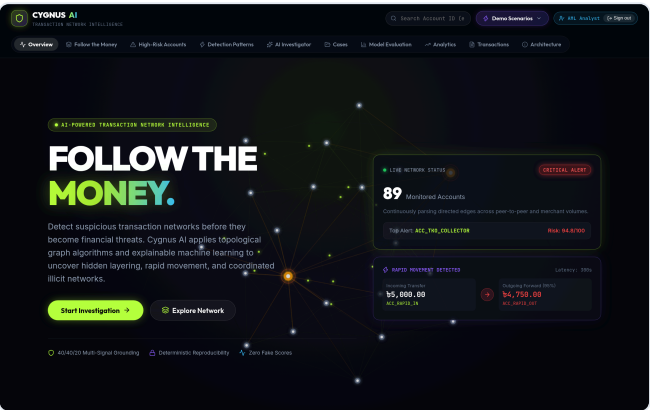
- **Live overview:** transactions, accounts, suspicious networks, elevated-risk accounts and average risk, computed from the real pipeline output.
- **Demo scenario bar:** one click jumps to each of the 11 planted scenarios: normal, fan-in, fan-out, rapid movement, chain, circular flow, coordinated network, structuring, the MFS mule ring, the hundi agent and the risk-area account takeover.
- **Follow the Money:** pick any account and expand its money network 1 or 2 hops; arrows show where money went next, colours show risk.
- **Risk score card:** the 0–100 score with its ML, graph and behaviour parts, inflow and outflow in BDT (₳), and patterns.
- **AI Investigator:** typology, summary, key findings, numbered next steps, linked accounts, a source badge (rule-based or Claude) and case-report export.
- **High-risk accounts, detection patterns, analytics charts and a transaction ledger** for broader review.
- **Upload your own data:** new transactions are validated and analysed by the same pipeline.



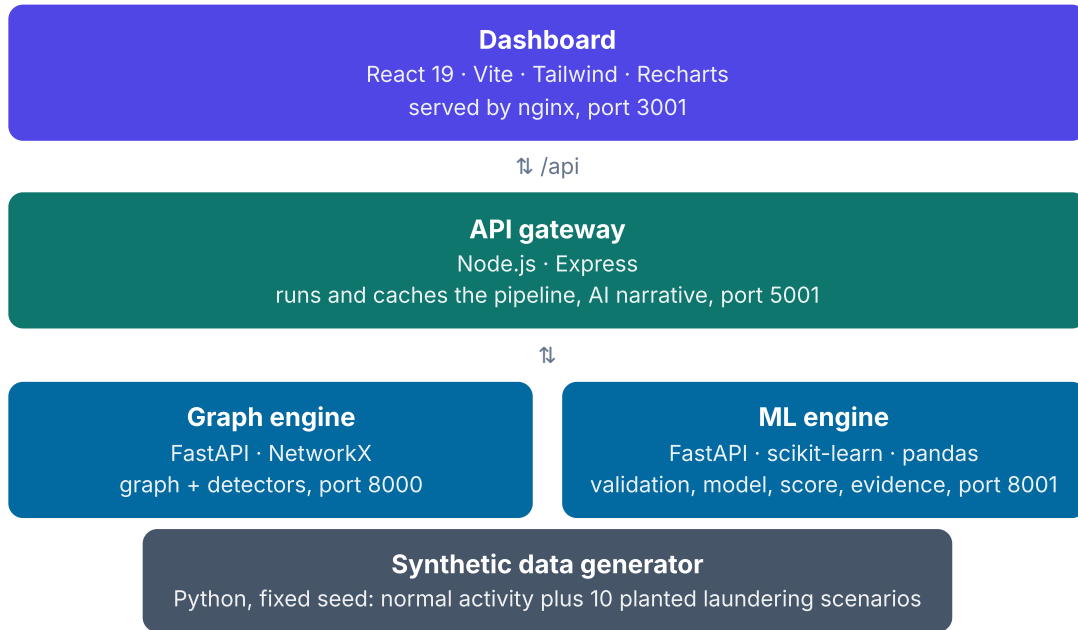
Follow the Money on the hundi agent: two funders top up ACC_HUNDI_AGENT, which pays the same six family wallets every cycle (score 84.0, HIGH). The mule ring (ACC_MULE_HUB, 92.3 CRITICAL) and the account-takeover collector (94.8 CRITICAL) are one click away in the scenario bar.



AI Investigator: typology, summary, key findings with real numbers, next steps and linked accounts.



Overview: headline numbers from the pipeline and the one-click demo scenario bar.



All four services start with one command (`docker-compose up --build`). The graph and ML engines also expose interactive API documentation at `/docs` , so another system can call them directly. The Node backend runs the same Python pipeline, so the dashboard and the APIs always agree.

LAYER	TECHNOLOGY	WHY WE CHOSE IT
Frontend	React 19, Vite, Tailwind CSS, Recharts, lucide icons	Fast, interactive, easy to demo
API	Node.js, Express	Simple REST gateway, file upload, caching
Graph	Python, NetworkX	Mature graph algorithms (cycles, components, paths)
ML	scikit-learn (Isolation Forest, RobustScaler), pandas, NumPy	Well-tested, explainable, runs on a laptop
Services	FastAPI, Uvicorn	Typed APIs with automatic docs
LLM (optional)	Anthropic Claude via the official <code>@anthropic-ai/sdk</code>	Plain-language case narrative, grounded in pipeline facts, with an offline fallback
Quality	pytest, benchmark script	Every detector and safety rule is tested
Deploy	Docker, Docker Compose, nginx	Same setup on a laptop and on the server

Main API endpoints

ENDPOINT	PURPOSE
GET <code>/api/stats</code>	Headline numbers for the dashboard
GET <code>/api/accounts</code> , <code>/api/accounts/:id</code>	Ranked accounts, one account's full score and evidence
GET <code>/api/network/:accountId</code>	The 1- or 2-hop money network around an account
GET <code>/api/patterns</code> , <code>/api/analytics</code> , <code>/api/transactions</code>	Pattern summary, charts, transaction ledger
GET <code>/api/investigate/:id</code>	AI Investigator briefing; Claude narrative when a key is set, rule-based otherwise

POST /api/pipeline/run	Analyse a new batch of uploaded transactions
GET /api/demo/scenarios	The one-click demo scenarios

11 Results, testing and safety

11.1 Results on the demo dataset

239 transactions, 0 rejected	89 accounts, 178 money links	15 suspicious networks (multi-account groups)	25.6 average risk score
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No normal account is flagged by any pattern (0 of 25), and the highest score of any normal account is 25.1 (LOW). Before the time-ordered loop check, ordinary peer-to-peer payments formed loops and were over-flagged.

Risk tiers across all accounts: **CRITICAL 5** **HIGH 5** **MEDIUM 16** **LOW 63**

11.2 Each planted scenario, and how the system scored it

The generator plants known laundering scenarios inside normal activity. This shows whether the system finds what it should.

SCENARIO	TARGET ACCOUNT	SCORE	TIER	PATTERNS FOUND	AI INVESTIGATOR TYPOLOGY
Normal Baseline Peer-to-Peer	ACC_NORM_004	2.5	LOW	none	No suspicious typology
Fan-In Aggregation Hub	ACC_FANIN_HUB	42.6	MEDIUM	fan in	Collection account
Fan-Out Dispersion Hub	ACC_FANOUT_HUB	44.9	MEDIUM	fan out	Dispersal account
Rapid Passthrough Movement	ACC_RAPID_MID	28.3	LOW	rapid movement	Pass-through account
Multi-Hop Transaction Chain	ACC_CHAIN_02	75.7	HIGH	transaction chain, rapid movement	Pass-through mule
Circular Layering Loop	ACC_CYCLE_B	60	MEDIUM	circular flow, rapid movement	Layering loop
Coordinated Account Cluster	ACC_COORD_00	41.2	MEDIUM	coordinated network	Coordinated account cluster
Structured Transfers	ACC_STRUCT_SRC	53.9	MEDIUM	structuring	Structuring (smurfing)
Mule-Ring (MFS Money-Mule Network)	ACC_MULE_HUB	92.3	CRITICAL	fan in, circular flow, transaction chain, rapid movement, structuring	Money-mule collection hub
Hundi Agent (Informal Remittance)	ACC_HUNDI_AGENT	84	HIGH	fan out, hundi operator	Hundi operator (informal remittance payout)
Risk-Area Account Takeover	ACC_TKO_COLLECTOR	94.8	CRITICAL	rapid movement, takeover collector	Account-takeover collector

11.3 Highest-risk accounts

ACCOUNT	SCORE	TIER	ML	GRAPH	BEHAVIOUR	PATTERNS
ACC_TK0_COLLECTOR	94.8	CRITICAL	98.5	100	77	rapid movement, takeover collector
ACC_MULE_HUB	92.3	CRITICAL	99.8	100	62	fan in, circular flow, transaction chain, rapid movement, structuring
ACC_MULE_01	92.2	CRITICAL	99.4	100	62	circular flow, transaction chain, rapid movement
ACC_MULE_02	92.2	CRITICAL	99.4	100	62	circular flow, transaction chain, rapid movement
ACC_MULE_03	92.2	CRITICAL	99.4	100	62	circular flow, transaction chain, rapid movement

11.4 Automated tests

TEST FILE	WHAT IT CHECKS	TESTS
tests/test_graph.py	Graph building, edge attributes, timestamps, pattern detectors, graph features	13
tests/test_ml.py	Behaviour features, Isolation Forest, score scaling, risk tiers, evidence text	8
tests/test_scenarios_benchmark.py	All 9 planted scenarios (including structuring and the mule ring), determinism, scores within 0-100	11
tests/test_security.py	Missing fields, bad IDs, self-transfers, bad amounts, NaN/Inf, bad timestamps, duplicates, empty data	12
tests/test_regressions.py	Scores identical across restarts; mixed time-zone timestamps do not crash	2
tests/test_upgrade.py	Structuring detector, out-of-order loops rejected, investigator briefings built from evidence	4
Evaluation, workflow and security tests	Evaluation, tier-aware scoring, roles, cases, audit log, PII masking, hundi and risk-area takeover detectors	21
API tests	Sign-in, roles, case actions and STR/SAR export over HTTP	10
Total, all passing		81

A separate benchmark (`evaluate_benchmark.py`) runs all 11 scenarios and checks that each target lands in its expected score band and that normal accounts stay under a false-positive ceiling. Results are deterministic: the same input always gives the same scores.

11.5 Responsible AI, privacy and security

CONCERN	WHAT WE DO
Privacy	Only synthetic data is used. No real customer, phone number or name appears anywhere. The model needs only IDs, amounts and times, not personal details.
Explainability	Every score is split into ML, graph and behaviour parts and comes with evidence sentences generated from computed metrics. No invented facts.

Fairness	The risk model never sees gender, age or district. Location is used only by the takeover detector, and only relative to each customer's own history (0 of 5,136 risk-area residents flagged). False alarms are checked by account type and district (Section 14).
LLM safety	The optional LLM receives only structured, computed facts about one account, never free text from transactions, so prompt injection through data has no path in. Scoring and decisions never depend on the LLM; it only rephrases, and the rule-based briefing is the fallback.
Human in the loop	Scores rank cases for review. They do not block accounts or decide guilt; an investigator makes the final call.
Input safety	Strict validation rejects malformed, impossible or duplicate records before they reach the graph or the model; covered by dedicated security tests.
Reproducibility	Fixed seeds and deterministic algorithms, so every result can be re-created for an audit.
Secrets	No passwords, keys or tokens are stored in the code repository.

Disclaimer. Detected patterns, scores and tiers are investigation aids on synthetic data. They do not prove financial crime, are not legal evidence, and are not official banking or regulatory thresholds.

848,752 synthetic Bangladesh-style MFS transactions with seven planted typologies and legitimate look-alikes. Trained on 2 networks, tuned on 1, tested on 3 unseen networks with different seeds (mean shown). Labels are exact by construction, so treat these as upper bounds for real upay data.

423,042

 labelled synthetic
transactions

39,490

accounts

1,010

 planted suspicious
accounts

**2 train · 1
tune · 3 test**

evaluation split

MODEL	PRECISION	RECALL	F1	PR-AUC	FALSE-POSITIVE RATE
Rules only (conventional monitoring)	26.5%	32.1%	0.29	0.20	2.35%
First prototype score (unsupervised)	9.3%	21.1%	0.13	0.10	5.40%
Isolation Forest (unsupervised)	8.2%	9.6%	0.09	0.09	2.83%
Graph pattern detectors	22.1%	82.9%	0.35	0.49	7.69%
Cygnus composite (current product)	69.8%	88.0%	0.78	0.84	1.00%
Gradient boosting, no graph features	80.3%	63.4%	0.71	0.76	0.41%
Random forest + graph features	90.7%	80.6%	0.85	0.92	0.22%
Gradient boosting + graph features	94.8%	83.8%	0.89	0.94	0.12%

What graph intelligence adds over transaction rules

On the same three held-out networks, transaction rules alone caught **324 of 1,010** suspicious accounts; the gradient-boosting model with graph features caught **846**. Of the 686 accounts the rules miss, the graph model catches 561. On evasive variants built to stay under rule thresholds, rules catch 9.1%, the model without graph features 66.1%, with graph features 82.6%. False alarms fall from 904 to 46.

Recall by typology

TYPOLGY	ACCOUNTS	RULES	GRAPH DETECTORS	ISOLATION FOREST	GBM, NO GRAPH	GBM + GRAPH
Mule ring with agent cash-out	264	34.8%	98.9%	14.8%	57.2%	92.0%
Digital hundi (informal remittance payout)	80	82.5%	93.8%	18.8%	78.8%	82.5%
Structuring under the threshold	204	32.4%	35.8%	0.0%	64.7%	67.2%
Circular layering	209	18.2%	99.5%	19.6%	49.8%	87.1%
Scam / betting collection account	60	40.0%	51.7%	1.7%	65.0%	61.7%
Low-and-slow mule chain	169	10.1%	97.6%	0.0%	75.1%	93.5%

Risk-area account takeover (collector)	24	87.5%	100.0%	4.2%	100.0%	95.8%
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Rules catch 82.5% of digital hundi because its amounts are regular; the graph detectors, including the new hundi detector in Section 7, catch 93.8%.

What each feature family adds (PR-AUC)

FEATURES	COUNT	PR-AUC
Transactional	21	0.660
+ temporal	35	0.817
+ graph and flow tracing	53	0.933
+ 2-hop neighbourhood	83	0.943

Feature importance (permutation: PR-AUC drop when one feature is shuffled)

FEATURE	TYPE	IMPORTANCE
average_transaction_amount	behaviour	0.041
amount_volatility	behaviour	0.035
chain_length_fast	graph	0.034
chain_length_slow	graph	0.023
total_outgoing	behaviour	0.018
nb_in_forward_share_slow	graph	0.017
forward_share_slow	graph	0.017
round_amount_share	behaviour	0.014

13 Analyst workflow and case management

Runs on our laptop today. The public server gets this version once the PR is merged.

STEP	WHAT THE ANALYST DOES	STATUS
Alert queue	Sign in as AML analyst; accounts ranked by risk tier	Built
Follow the Money	Expand 1 or 2 hops of the money network	Built
AI Investigator	Typology, evidence and next steps	Built
Open a case	Notes, status and escalation to the compliance officer	Built
STR / SAR draft	Exportable report built from the case evidence	Built
Audit trail	Hash-chained log of case actions and PII reveals	Built

Demo script. Sign in as an AML analyst, open the top CRITICAL alert, follow the money two hops, read the AI Investigator, open a case and add a note, escalate it, export the STR/SAR draft, then show the audit log and the Model Evaluation tab.

14 Scalability, integration and security controls

Measured throughput

1,061,510

transactions scored in
18.9 s

~56,000 / s

scoring throughput, one
process

< 6 ms

API response time, p95
(every endpoint)

0.21 s

re-score the last 24 hours
(9,508 tx)

8-core laptop, Python 3.11, one process in Docker, no GPU, peak 1.9 GB. Full pipeline with graph and evidence text: ~8,600 tx/s.

Target. All Bangladesh MFS together run about 22 million transfers a day, roughly 250 a second on average [2]. The batch scoring path already runs about 56,000 a second in one laptop process. What production adds is real-time streaming and multi-hop queries at scale, which the target architecture below is designed for.

LAYER (ROADMAP)	DESIGN
Stream in	Kafka topic fed by upay's core ledger or AML feed (change data capture), so scoring runs as transfers happen
Window	Incremental graph features over sliding 1-hour, 24-hour and 7-day windows instead of rebuilding the whole graph
Graph store	Neo4j or TigerGraph for multi-hop queries on hundreds of millions of edges
Score	Current detectors and models, plus a temporal graph neural network once analyst labels exist, behind the same API
Integrate	Cases and STR drafts pushed to upay's AML / case-management system through a documented REST contract

Security and responsible-AI controls built in the prototype

CONTROL	WHAT IT DOES
Strict input validation	bad amounts, NaN/Inf, bad timestamps, self-transfers, duplicates rejected
Role-based access	sign-in as AML analyst or compliance officer
Hash-chained audit log	case actions and data reveals are chained, so tampering shows
PII masking	personal details masked by default; every reveal is logged
Bias check	legit agents flagged 98.7% (first prototype) → 0.57% now; no district effect (p = 0.30)
Grounded explanations	every sentence quotes a computed metric; the optional LLM only rephrases
Human in the loop	scores prioritise review; analysts decide and file STRs

Bias check across account tiers

TIER	LEGITIMATE ACCOUNTS	RULES	FIRST PROTOTYPE SCORE	CURRENT CYGNUS SCORE	GBM + GRAPH
Personal	35,013	1.96%	0.43%	1.08%	0.11%
Agent	1,397	5.44%	98.71%	0.57%	0.57%
Merchant	2,070	6.91%	26.43%	0.00%	0.00%

False-positive rate per account tier on legitimate accounts. District is never a model input; by district (14 districts) the model's false-positive rate ranges 0.00% to 0.24% with chi-square p = 0.298, no evidence of dependence. Our first prototype score

flagged almost every legitimate agent, because collecting from many customers is what an agent does; the current score is tier-aware.

Production roadmap for security and governance

- SSO with upay identity and MFA for analysts
- TLS everywhere and encryption at rest for case data
- Write-once audit storage and regular external penetration tests
- Model-drift and alert-volume monitoring with monthly threshold review
- Privacy governance: data minimisation, retention limits, access reviews
- Fairness review on real data across regions, account types and agent tiers
- Cross-provider analytics without sharing raw PII (federated learning or secure aggregation)

15 Impact and business value

For MFS providers like upay

Spot mule networks before the money is cashed out. Fewer fraud losses, fewer chargeback and complaint cases, and stronger AML/CFT reporting to the regulator.

For compliance analysts

See a whole multi-hop flow in one click instead of tracing it row by row. A ranked queue with reasons, so the riskiest cases are handled first.

For wallet users

Fewer people recruited as mules or losing money to scams, and more trust in digital money, which drives adoption.

Measured on the benchmark

73.6%

fewer alerts at equal recall

324 alerts instead of 1,228 to find the same 324 accounts

904 → 46

false alarms
on 39,490 held-out accounts

32% → 84%

suspicious accounts caught
rules vs graph model, same networks

76 → 21 min

analyst time per true case
at 20 minutes per alert (assumed)

Modelled savings (assumptions stated; to be measured in the pilot)

- Assumption: 20 minutes of analyst review per alert. The pilot replaces it with upay's measured time.
- Rules: 1,228 alerts = 409 analyst hours, finding 324 suspicious accounts.
- Graph model: 892 alerts = 297 analyst hours, finding 846.
- **Same 324 accounts with the graph model: 324 alerts, 108 hours instead of 409.**
- Synthetic networks are ~2.5% suspicious, far above production, so real precision will be lower.

Proposed 8-week pilot with upay

- Shadow mode on governed upay data: Cygnus scores, analysts work as usual
- Analysts label every alert; we measure precision@100 and recall on confirmed cases
- Time each case with and without Cygnus to get real minutes saved
- Calibrate thresholds and tiers with upay's AML team
- Exit report: ROI, false-positive cost and a go / no-go for production

Business model

- **B2B SaaS licence** for MFS providers, banks and fintechs: the detection engine plus the investigator dashboard.

- **API pricing per volume** for institutions that want to call the scoring pipeline from their own monitoring systems.
- **Integration and tuning services:** mapping a provider's data to the five-field input and tuning thresholds with their analysts.

Why it can scale

- The input is just five common fields, so connecting a new data source is a mapping exercise, not a rebuild.
- Graph and ML engines are separate services with their own APIs, so each can be scaled or replaced on its own.
- Unsupervised learning works before any fraud labels exist; analyst decisions can later be added as labels to train a supervised model.
- For millions of transactions per day, the same design moves to streaming ingestion and a graph database (see roadmap).

16 Honest limitations and roadmap

What a prototype cannot do yet

- All data is synthetic, and the simulator and detectors were built by the same team, so the measured numbers are an upper bound for real upay traffic.
- Suspicious accounts are about 2.5% of each test network, far above production, so precision on real data will be lower at the same threshold.
- Labels are exact by construction. Real labels (filed STRs, confirmed fraud) are delayed, incomplete and noisy.
- Neighbourhood features are fixed 2-hop averages, not a trained graph neural network; a learned GNN needs real labels first.
- Analyst-hour figures assume 20 minutes per alert; the pilot will measure the real figure.
- Throughput was measured in one process on a laptop. Kafka streaming and a graph database are designed, not built.
- The on-site demo also runs on the team laptop at <http://localhost:3001> as a backup to the live server.

Roadmap

STAGE	GOAL
Stage 1 · Done	Prototype: Seven detectors, Isolation Forest, explainable score, AI Investigator
Stage 2 · Done	Analyst platform: 849k-transaction labelled benchmark, trained model, cases, roles, audit log, takeover and hundi detectors
Stage 3 · Next	upay pilot: Governed upay data in shadow mode, analyst feedback, calibrated tiers
Stage 4 · Future	MFS scale: Streaming ingest, graph database, temporal GNN, cross-provider privacy-preserving analytics

17 How to run it

```
git clone https://github.com/ManamiMayoki/CPC_DIU_HACKATHON_2026
cd CPC_DIU_HACKATHON_2026
docker-compose up --build
# Dashboard      http://localhost:3001
# API            http://localhost:5001/api/health
# Graph engine   http://localhost:8000/docs
# ML engine      http://localhost:8001/docs
```

Without Docker: `pip install networkx pandas numpy scipy scikit-learn pytest` , then `python ml/inference.py` runs the full pipeline, `python -m pytest tests/` runs the tests and `python evaluate_benchmark.py` runs the scenario benchmark. `python data/synthetic/generator.py` regenerates the synthetic data. No environment variables are required. Optionally set `ANTHROPIC_API_KEY` before `docker-compose up` to enable the Claude narrative; the key is never stored in the repository.

Repository map

frontend/	React dashboard (Follow the Money, AI Investigator, analytics)
backend/	Express API and the adapter that runs the Python pipeline
graph-engine/	Graph construction, pattern detectors, graph features, FastAPI service
ml/	Validation, behaviour features, Isolation Forest, risk scoring, evidence, FastAPI service
data/synthetic/	Synthetic data generator and sample dataset
tests/	Unit, integration, benchmark, security and regression tests
docs/	Architecture and API contract

External resources used

Open-source libraries only: NetworkX, scikit-learn, pandas, NumPy, FastAPI, Uvicorn, pytest, React, Vite, Tailwind CSS, Recharts, lucide-react, Express, nginx and Docker. Optional: the Anthropic Claude API for the investigator narrative (off unless a key is provided). All transaction data is synthetic and generated by our own script. No external dataset is used.

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